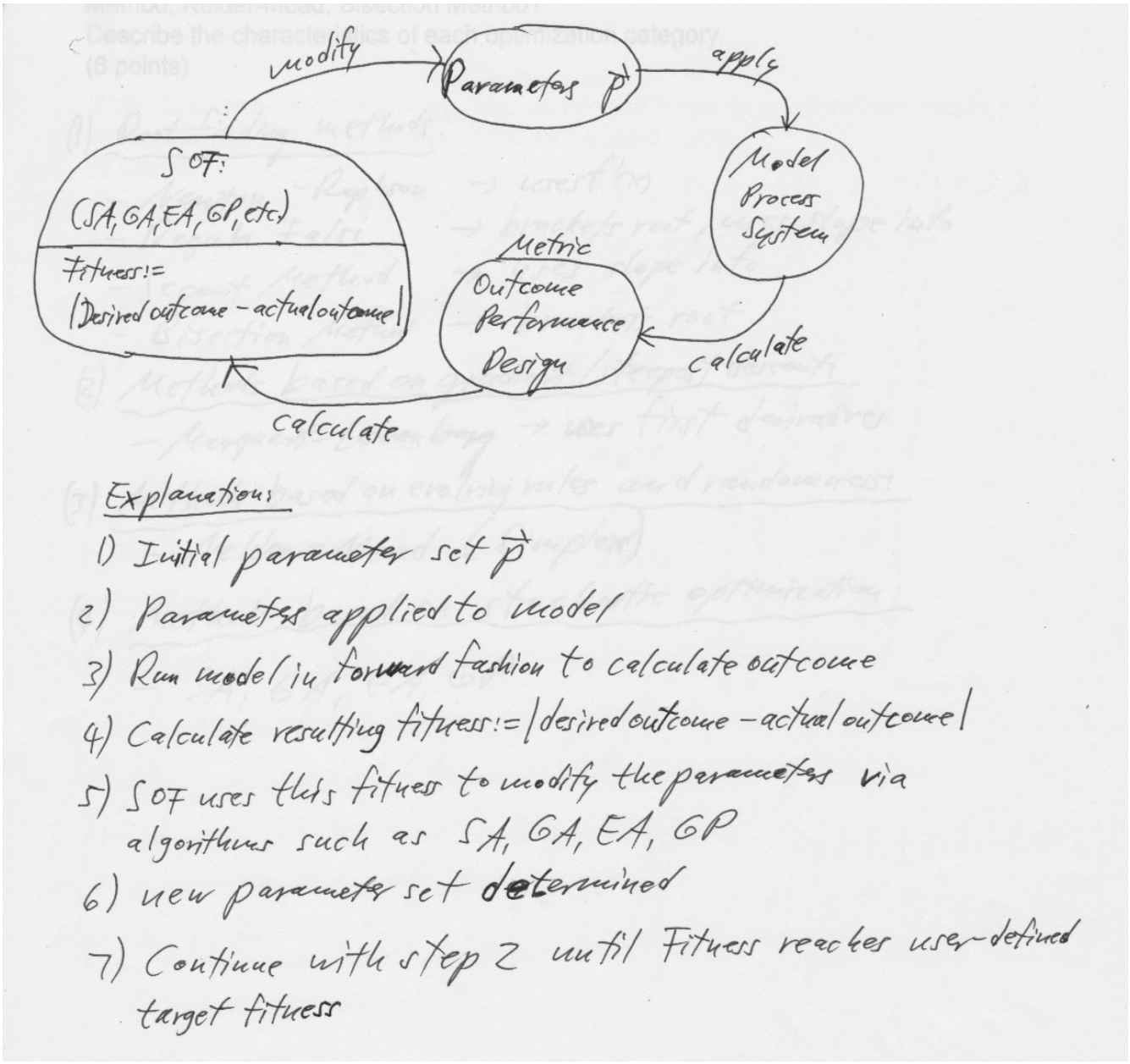
Questions for Final Exam ECE 330:

Name: _____

Question #1 (6 points total):

Describe the Stochastic Optimization Framework in detail (i.e., graphical sketch with labeling of all components and written explanation of how it works).
(4 points)



Explanations:

1) Initial parameter set $\vec{p}$
2) Parameters applied to model
3) Run model in forward fashion to calculate outcome
4) Calculate resulting fitness!=|desired outcome - actual outcome|
5) SOF uses this fitness to modify the parameters via algorithms such as SA, GA, EA, GP
6) new parameter set determined
7) Continue with step 2 until Fitness reaches user-defined target fitness

Question #2 (16 points total):

a) Write down mathematically the Gauss-Legendre Integration (GLI) approximation of an integral of a function f(x) over an arbitrary finite interval [a,b]. Explain at what locations the function f(x) is being evaluated in the GLI approximation.
(5 points)

b) If you had to integrate a polynomial $P_{330}(x)$ of degree 330, what would be the lowest degree of a Legendre polynomial to get an exact result with GLI and why?
(3 points)

c) Graphically describe the Trapezoidal Rule and explain it. List the mathematical approximation formula for the integral over several sub-intervals.
(4 points)

d) Graphically describe the Simpson Rule and explain it. List the mathematical approximation formula for the integral over several sub-intervals.
(4 points)

(4 points)

a) $\int_a^b f(u)\,du = \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{(b-a)x}{2}+\frac{b+a}{2}\right)dx \overset{\text{GLI}}{\approx} \frac{b-a}{2}\sum_{i=1}^{n} w_i f\left(\frac{(b-a)x_i}{2}+\frac{b+a}{2}\right)$

with $x_{i=1,\dots,n}$ = n roots of Legendre-polynomial L_n in $(-1,1)$ (n: degree)

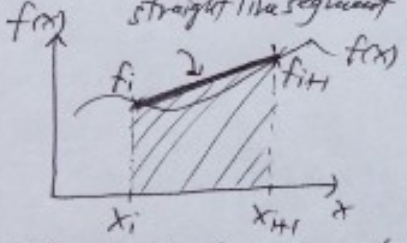
and w_i = integration weights, $f(x)$ = integrand

b) all polynomials up to degree $(2n-1)$ are integrated exactly over $(-1,1)$ using GLI (i.e. Legendre-polynomial of degree n) $\Rightarrow 2n-1=330$

$\Rightarrow n=\frac{331}{2}$

$\Rightarrow n=166$ (n is integer)

c) Trapezoidal Rule (TR)

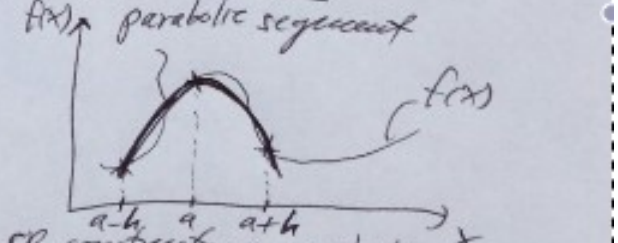


TR approximates integral of function f(x) between each pair of neighboring points x_i, x_{i+1} by the area under the connecting straight line.

$I_{TR} = \frac{h}{2}f_1 + hf_2 + hf_3 + \dots + hf_{n-1} + \frac{h}{2}f_n$

or $I_T = \frac{1}{2}(x_{i+1}-x_i)(f_{i+1}+f_i) + \frac{1}{2}(x_{i+2}-x_{i+1})(f_{i+2}+f_{i+1}) + \dots + \frac{1}{2}(x_n-x_{n-1})(f_n-f_{n-1})$

d) Simpson Rule (SR)



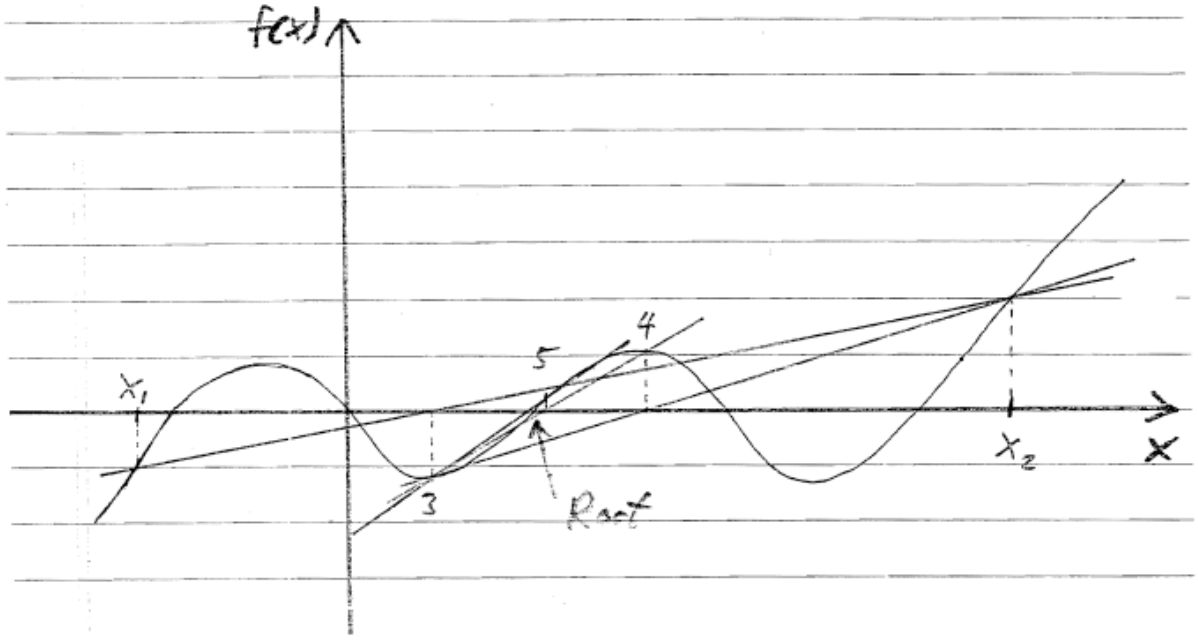
SR constructs a parabolic arc through each triad of points x_i, x_{i+1}, x_{i+2} and integrates over the areas underneath the respective parabolic arcs.

$I_{SR} = \frac{h}{3}f_1 + \frac{4h}{3}f_2 + \frac{2h}{3}f_3 + \frac{4h}{3}f_4 + \dots + \frac{4h}{3}f_{n-1} + \frac{h}{3}f_n$

Question #3 (6 points total):

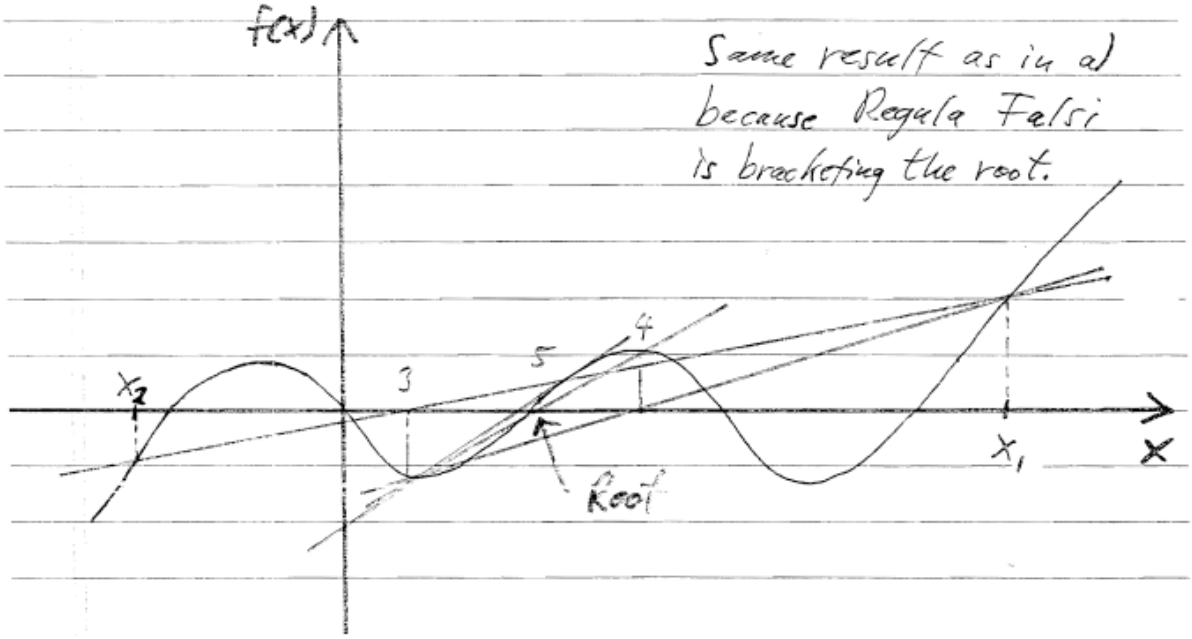
a) Applying the Regula Falsi with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!

(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the respective results from a and b.

(3 points)



Question #4 (10 points total):

Describe the Simulated Annealing (SA) algorithm in detailed pseudo-code form. (10 points)

```
Set Temperature T = T0
Initial parameter vector P
Calculate initial current fitness/energy E_current
Set best fitness/energy E_best = E_current
Set best parameter vector P_best = P
Set cooling rate: 0 < lambda < 1

LOOP:
{
    Modify parameter vector P to yield P'
    Calculate resulting temporary fitness/energy: E_temp

    if (E_temp< E_current)
    {
        E_current = E_temp
        make P' the new P: P = P'
        if (E_current < E_best)
        {
            E_best = E_current
            P_best = P
            if (E_best < user-defined fitness/energy)
            {
                STOP optimization
            }
        }
    }
    else
    {
        if (random number [0...1] < exp[-(E_temp - E_best)/T])      [COMMENT: accept unfavorable step with Boltzmann-probability [0...1]]
        {
            E_current = E_temp
            make P' the new P: P = P'
        }
        else
        {
            reject P' and go back to P
        }
    }
    T = lambda * T      [COMMENT: cool temperature for Boltzmann-probability, i.e., make it harder to accept unfavorable steps]
    if (T < user-defined minimum T)
    {
        STOP optimization
    }
    else
    {
        goto LOOP
    }
}
```

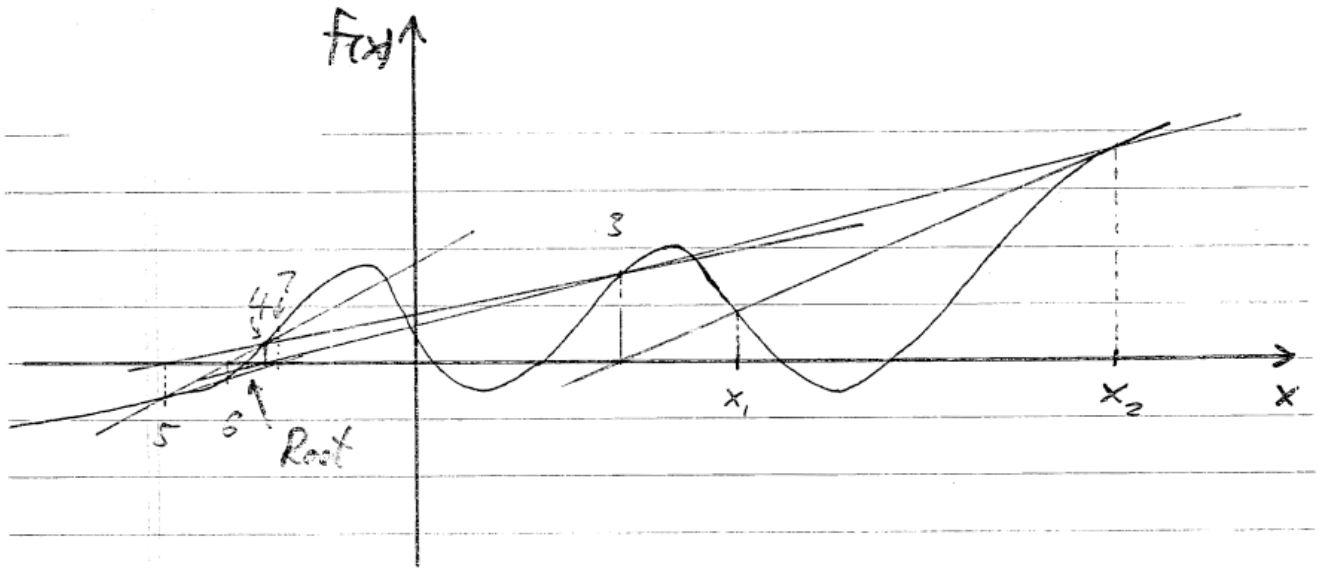
} exit criterion (I)

} exit criterion (II)

can also be max. number of allowed/prescribed iterations exhausted/exceeded

Question #5 (6 points total):

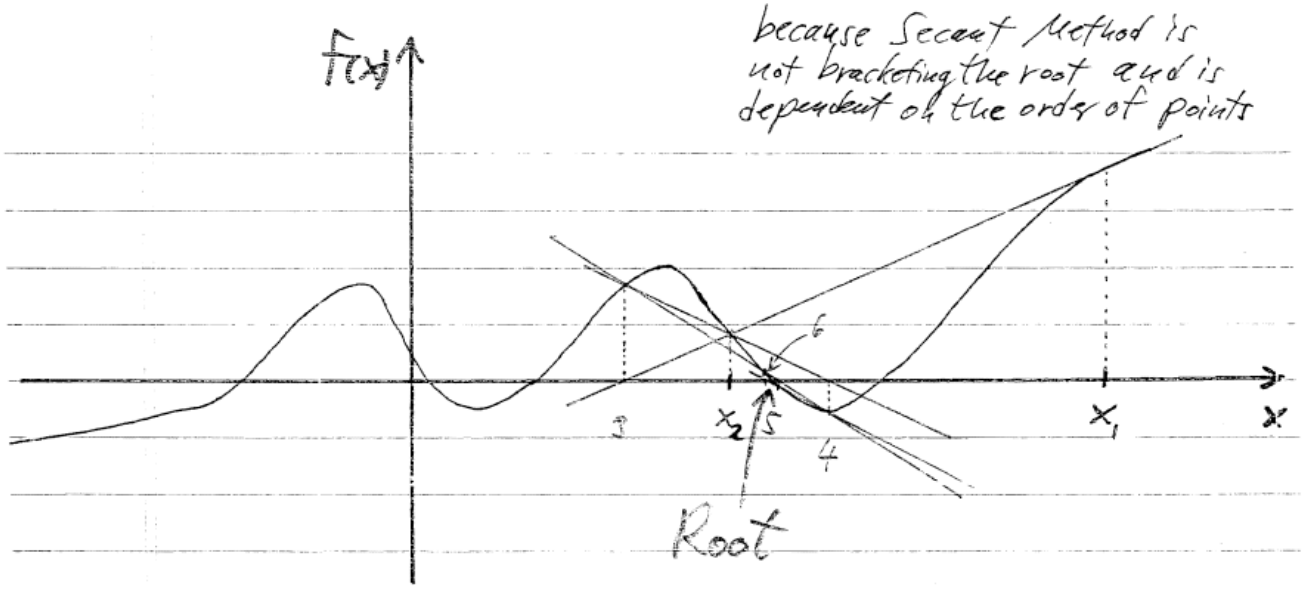
a) Applying the Secant Method with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!
(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the respective results from a and b.
(3 points)

(5 points)

Result is different from a) because Secant Method is not bracketing the root and is dependent on the order of points



Question #6 (10 points total):

a) You have the following parameter vector in the population of a Genetic Algorithm:
$[1 \quad 3 \quad -1 \quad 9 \quad 4 \quad -5 \quad -7 \quad 8]$
Apply the *inversion* operator and write down the resulting parameter vector.
(2 points)

b) The following two children have resulted from a *crossover* operation in a Genetic Algorithm:

Child 1: $[2 \quad -3 \quad 1 \quad -1 \quad 5 \quad 3 \quad -7 \quad -8]$

Child 2: $[8 \quad 2 \quad 7 \quad 6 \quad 4 \quad 5 \quad -9 \quad 0]$

The crossover area started at position 4 (including) and ended at position 6 (including). Numbering starts with 1. What were the initial parent parameter vectors?
(4 points)

c) The following two parents have been selected for a *crossover* operation in a Genetic Algorithm for a Traveling Salesman Problem:

Parent 1: $[1 \quad 3 \quad 5 \quad 2 \quad 7 \quad 6 \quad 4 \quad 8]$

Parent 2: $[8 \quad 2 \quad 7 \quad 1 \quad 6 \quad 3 \quad 4 \quad 5]$

The crossover area starts at position 3 (including) and ends at position 6 (including). Numbering starts with 1. Which two resulting offspring parameter vectors will result? Will there be a problem with the offspring parameter vectors in this case or not? Justify your answer in any case.
(4 points)

a) $[8 \quad -7 \quad -5 \quad 4 \quad 9 \quad -1 \quad 3 \quad 1]$

b) Child 1: $[2 \quad -3 \quad 1 \quad -1 \quad 5 \quad 3 \quad -7 \quad -8]$
[----------]

Child 2: $[8 \quad 2 \quad 7 \quad 6 \quad 4 \quad 5 \quad -9 \quad 0]$
[----------]

Parent 1: $[2 \quad -3 \quad 1 \quad 6 \quad 4 \quad 5 \quad -7 \quad -8]$

Parent 2: $[8 \quad 2 \quad 7 \quad -1 \quad 5 \quad 3 \quad -9 \quad 0]$

c) Parent 1: $[1 \quad 3 \quad 5 \quad 2 \quad 7 \quad 6 \quad 4 \quad 8]$
[----------]

Parent 2: $[8 \quad 2 \quad 7 \quad 1 \quad 6 \quad 3 \quad 4 \quad 5]$
[----------]

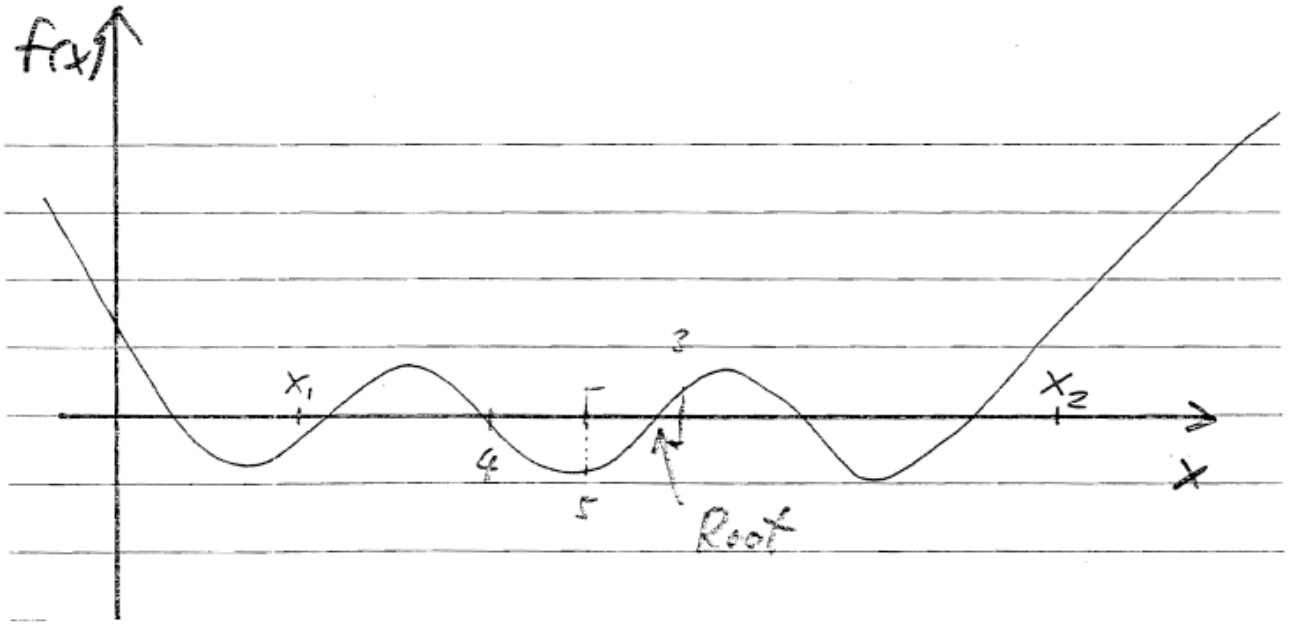
Child 1: $[1 \quad 3 \quad 7 \quad 1 \quad 6 \quad 3 \quad 4 \quad 8]$

Child 2: $[8 \quad 2 \quad 5 \quad 2 \quad 7 \quad 6 \quad 4 \quad 5]$

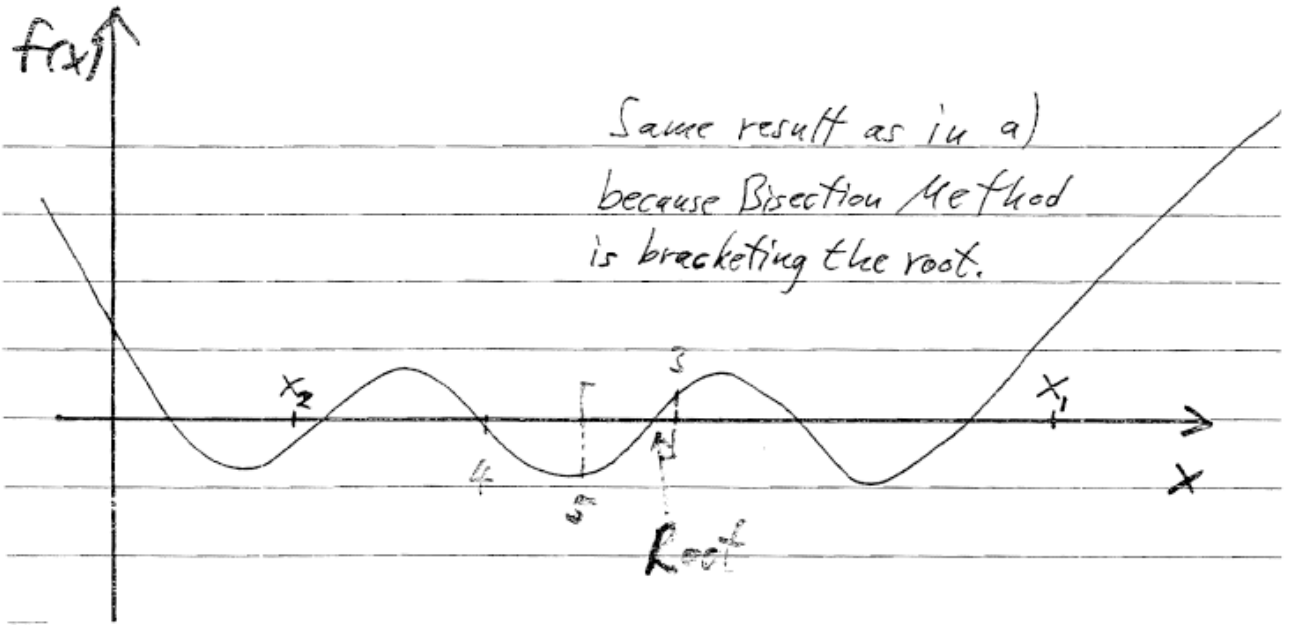
Both offspring parameter vectors (children) are invalid for the TSP problem because of both duplicate and missing city numbers.

Question #7 (6 points total):

a) Applying the Bisection Method with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!
(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the result from a and b.
(3 points)



Question #8 (10 points total):

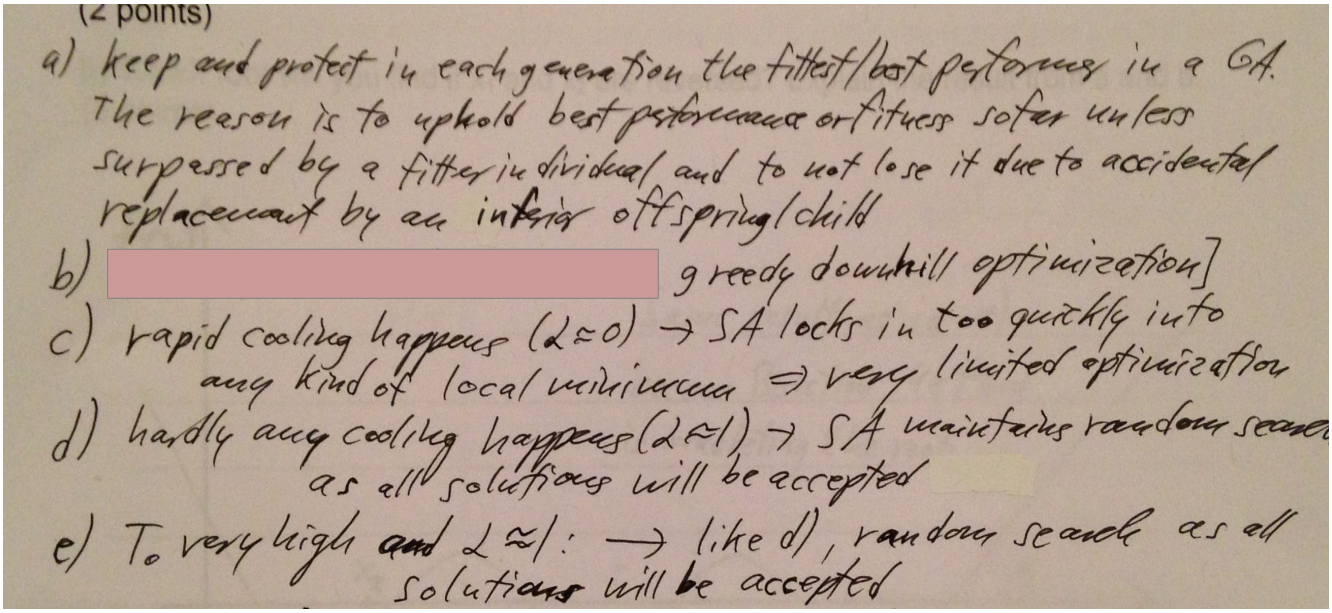
a) Explain the concept of the Elitist approach in Genetic Algorithms. Why is it important?
(2 points)

b) What does Simulated Annealing turn into when you eliminate the section of the algorithm that accepts a worse performing solution via the Boltzmann-probability?
(2 point)

c) What happens to Simulated Annealing when the cooling rate Lambda is very close to 0?
(2 points)

d) What happens to Simulated Annealing when the cooling rate Lambda is very close to 1?
(2 points)

e) What happens to Simulated Annealing when the initial temperature is very high and the cooling rate Lambda very close to 1?
(2 points)


(2 points)
a) keep and protect in each generation the fittest/best performer in a GA. The reason is to uphold best performance or fitness so far unless surpassed by a fitter individual and to not lose it due to accidental replacement by an inferior offspring/child
b) [redacted] greedy downhill optimization]
c) rapid cooling happens ($\lambda \approx 0$) → SA locks in too quickly into any kind of local minimum ⇒ very limited optimization
d) hardly any cooling happens ($\lambda \approx 1$) → SA maintains random search as all solutions will be accepted
e) T_0 very high and $\lambda \approx 1$: → like d), random search as all solutions will be accepted

Question #9 (10 points total):

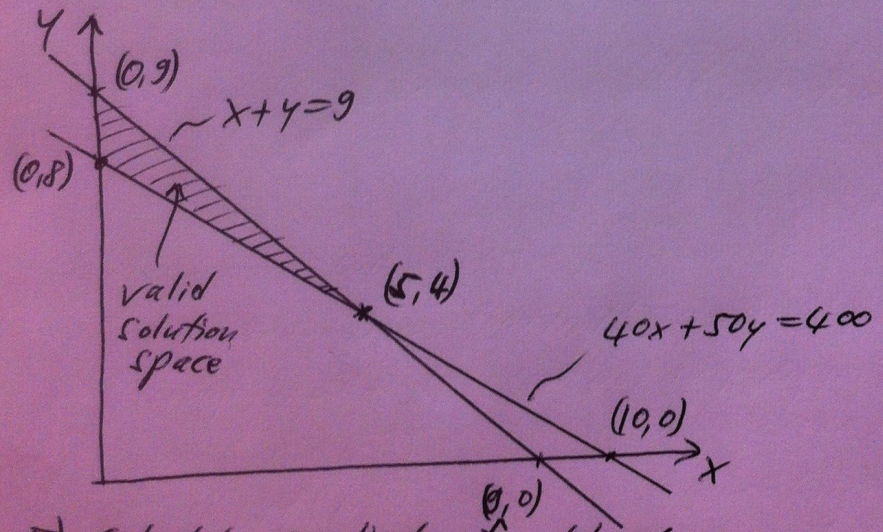
A company is preparing a trip for 400 employees. The transportation service has 10 buses of 50 seats each and 8 buses of 40 seats, but only 9 drivers available. The rental cost for a large bus is $800 and $600 for the small bus. Calculate how many buses of each type should be used for the trip for the least possible cost.
(10 points)

1) define unknowns:
x = # small buses
y = # big buses

2) Write objective function:
$f(x,y) = 600x + 800y$

3) Write constraints as system of inequalities:
$40x + 50y \geq 400$
$x + y \leq 9$
$x \geq 0$ $x \in \mathbb{N}_0$ (integers)
$y \geq 0$ $y \in \mathbb{N}_0$ (integers)

4) Find set of feasible solutions:



5) Calculate coordinates of valid vertices:
(I): $x+y=9 \Rightarrow (0,9)$
(II): $40x+50y=400 \Rightarrow (0,8)$
(III): $\left.\begin{matrix} x+y=9 \\ 40x+50y=400 \end{matrix}\right\} \Rightarrow (5,4)$

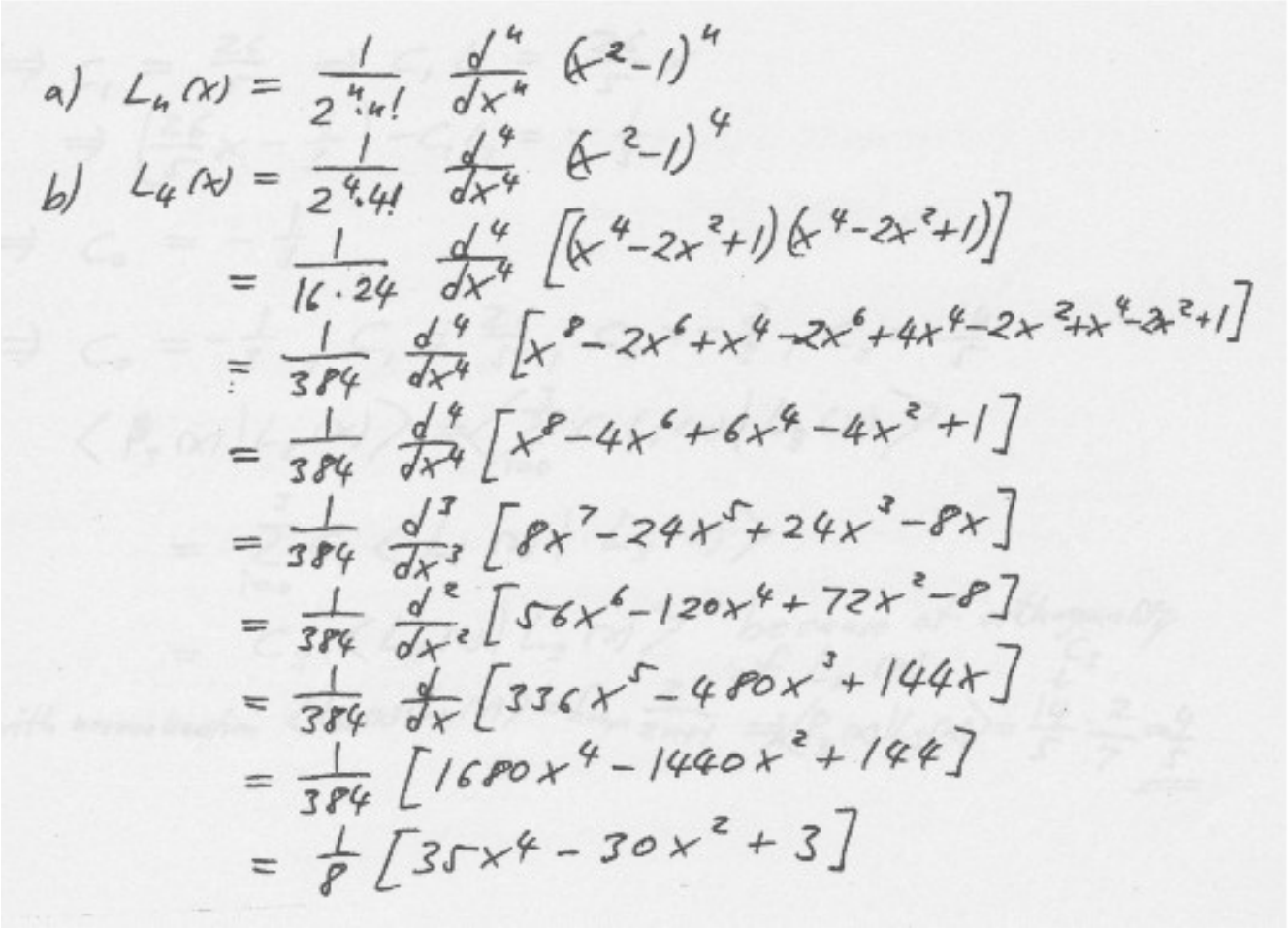
6) Calculate objective function value at each vertex:
$f(0,9) = 7200$
$f(0,8) = 6400$
$\boxed{f(5,4) = 6200}$

$\Rightarrow$ minimum cost achieved with 5 small and 4 large buses

Question #10 (8 points total):

a) List Rodrigues formula for Legendre Polynomials.
(2 points)

b) For the polynomial $P_4(x)=\frac{35}{8}x^4-\frac{35}{4}x^3-17x^2-330x+\frac{49}{37}$: calculate the explicit value of $\langle P_4(x)|L_4(x)\rangle$.
(6 points)



So, if $P_4(x)$ were to be expanded into Legendre polynomials, the coefficient for the x^4-term would be $C_4 = 1$.

Therefore, $\langle P_4(x)|L_4(x)\rangle$ = $\langle L_4(x)|L_4(x)\rangle$ because of orthogonality

and $\langle L_4(x)|L_4(x)\rangle$ = $\frac{2}{2n+1}$ = $\frac{2}{9}$

Question #11 (6 points total):

a) What happens to the time development of an oscillator with no driving force term if the initial conditions for the oscillator are: location x(t=0) = 0 and velocity v(t=0) = 0? (2 points)

b) The following differential equation is given: $\frac{dy}{dt} = Ay^2 - Bt^3$ with $A, B = const.$

Derive explicitly the update equation for integrating this differential equation using the Runge-Kutta-4 Method.
(4 points)

a) because of no driving force term the system will remain at rest, i.e., will NOT start oscillating.

b)

$$\frac{dy}{dt} = f(y,t) \qquad \frac{dy}{dt} = Ay^2 - Bt^3 = f(y,t)$$

$$k^{(1)} = f(y,t) \qquad \Rightarrow k^{(1)} = Ay^2 - Bt^3$$

$$k^{(2)} = f\left(y + \frac{h}{2}k^{(1)}, t + \frac{h}{2}\right) \qquad \Rightarrow k^{(2)} = A\left[y + \frac{h}{2}k^{(1)}\right]^2 - B\left(t + \frac{h}{2}\right)^3$$

$$k^{(3)} = f\left(y + \frac{h}{2}k^{(2)}, t + \frac{h}{2}\right) \qquad \Rightarrow k^{(3)} = A\left[y + \frac{h}{2}k^{(2)}\right]^2 - B\left(t + \frac{h}{2}\right)^3$$

$$k^{(4)} = f\left(y + hk^{(3)}, t + h\right) \qquad \Rightarrow k^{(4)} = A\left[y + hk^{(3)}\right]^2 - B(t+h)^3$$

$$\Rightarrow y(t+h) = y(t) + h\left[\frac{k^{(1)}}{6} + \frac{k^{(2)}}{3} + \frac{k^{(3)}}{3} + \frac{k^{(4)}}{6}\right]$$

Question #12 (6 points total):

Approximate the general Laplace equation in cylindrical coordinates:

$$\Delta\phi(z,r,t) = \frac{\partial^2\phi}{\partial z^2} + \frac{1}{r}\frac{\partial\phi}{\partial r} + \frac{\partial^2\phi}{\partial r^2} + \frac{1}{r^2}\frac{\partial^2\phi}{\partial t^2} = 0$$

with the improved Two-Point and Three-Point Formulas and ignore the respective leading error terms. Mathematically reduce the result as much as possible.
(6 points)

$$\Delta\phi \approx \frac{\phi(z+h,r,t)+\phi(z-h,r,t)-2\phi(z,r,t)}{h^2}$$

$$+\frac{1}{2r}\,\frac{\phi(z,r+h,t)-\phi(z,r-h,t)}{h}$$

$$+\frac{\phi(z,r+h,t)+\phi(z,r-h,t)-2\phi(z,r,t)}{h^2}$$

$$+\frac{1}{r^2}\,\frac{\phi(z,r,t+h)+\phi(z,r,t-h)-2\phi(z,r,t)}{h^2} = 0 \quad \Big| \cdot h^2$$

$$\Rightarrow \phi(z+h,r,t)+\phi(z-h,r,t)+\phi(z,r+h,t)+\phi(z,r-h,t)$$

$$-4\phi(z,r,t)+\frac{h}{2r}\left[\phi(z,r+h,t)-\phi(z,r-h,t)\right]$$

$$+\frac{1}{r^2}\left[\phi(z,r,t+h)+\phi(z,r,t-h)-2\phi(z,r,t)\right]$$

$$= \phi(z+h,r,t)+\phi(z-h,r,t)+\left(1+\frac{h}{2r}\right)\phi(z,r+h,t)$$

$$+\left(1-\frac{h}{2r}\right)\phi(z,r-h,t)-\left(4+\frac{2}{r^2}\right)\phi(z,r,t)$$

$$+\frac{1}{r^2}\left[\phi(z,r,t+h)+\phi(z,r,t-h)\right] = 0$$